

Understanding Financial Asset Price Dynamics Through Structural Shocks^{*}

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Abstract

This paper introduces an extended Bayesian Vector Autoregressive (BVAR) model to analyse macroeconomic shocks affecting the euro area (EA) from 2007 to 2024. The model incorporates a newly constructed Energy Price Index (EPI) and identifies energy supply shocks alongside more conventional macroeconomic drivers such as demand and supply shocks as well as both conventional and unconventional monetary policy shocks. Using sign and magnitude restrictions, we decompose structural drivers across eight variables. The results provide insights into the dynamics of inflation expectations, energy price shocks, and monetary policy effectiveness in the euro area.

1 Introduction

The global economy has experienced a significant rise in inflation rates following the COVID-19 pandemic. Pinpointing this inflationary trend to a single source is challenging due to the confluence of multiple factors. Notably, some analysts suggest that expansive monetary policies and robust fiscal stimulus measures have rekindled demand-side pressures across various regions. Concurrently, disruptions in supply chains and substantial rises in commodity prices are also perceived as critical factors elevating inflation levels. For policymakers, discerning the relative impact of demand versus supply influences is essential to devise effective policy measures. In our study, we introduce a methodology that leverages daily financial data to delineate both domestic and global structural determinants of financial asset prices, thereby providing clearer insights into the mechanisms influencing current economic conditions.

Our modelling framework utilises a two-country Bayesian Vector Autoregressive (BVAR) model for the euro area and the United States. We identify the structural drivers of daily asset price changes in these economies by applying sign and magnitude restrictions, a standard method in recent empirical macroeconomic research [Arias et al. \(2018\)](#). Building on the work of [Brandt et al. \(2026\)](#), we broaden the set of asset prices by including the Energy Price Index (EPI) and Inflation Linked Swaps (ILS) rates. This addition allows us to identify demand and supply shocks in the euro area, as well as a global energy supply shock.

The model also includes both conventional and unconventional monetary policy shocks, providing a more nuanced understanding of the inflation dynamics in the euro area. By decomposing the sources of inflation expectations, our findings offer insights into the period from 2007 to 2024, focusing particularly on the post-pandemic inflation surge and energy price volatility.

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